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Areas where simulation is making a significant mark

“A lot of new fields have also begun actively using simulation, like computational finance, computational biology and energy production,” says Rao. Electromagnetics plays a very important role in product development. Simulation solutions give engineers the power to make decisions related to electromagnetic variations in various designs.

Next is high-frequency product design. “For RF and microwave design engineers, product integrity, system integrity and signal integrity are very critical areas of decision making. Moving on to low-frequency electronics areas like motors, turbines and other areas where you have to do system-level design and then break it down into a variety of components that come together to make a low-frequency product, decisions have to be taken to

make it more efficient and provide the functionality that it is designed for,” adds Dutta.

Coming to automobiles, we all know that today’s automobiles have more electronics than some home appliances. Electric and hybrid vehicles are the future and that is where you have the battery balanced with the motor connected using a power electronic circuit to work on them efficiently. All these pieces need to be designed in an optimal manner keeping in mind thermal and electromagnetic phenomena that take place inside.

Cooling is one of the biggest concerns when developing an electronic device.

“You don’t want the device to overheat. That is one area where we help a lot. With our product, you can

see exactly how each component heats up and how all of these heat up the product when acting together and also what might happen when such overheating occurs, including development of stress points resulting in eventual breakdown. It can also help to identify accurate ways of cooling down the system. It can tell you if you need forced convection or some novel methods like phase change materials to reach the same purpose,” adds Dravid.

Simulation tools can model the spatial flow of current through a chip as well as induction effects, eddy-current effects and other spatial phenomena, helping you to find out how these can affect both the component and the device.

With embedded software defining the functionality of most of the electronic hardware used in automotive, aerospace, consumer electronics and medical equipment, flawless functioning of these systems becomes all the more critical.

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“Hardware-in-loop simulation provides the best balance ensuring quality through exhaustive verification and cost-effectiveness. This helps today’s system builders to reduce time to market while still ensuring reliability. The trend that we are seeing is percolation of this hardware-in-loop testing into the RF system design area for validation” — Satish Mohanram, Technical Marketing Manager, National Instruments-India